

structure, and manufacturing. The increased data speed and network capacity, much lower latency, and faster round-trip data transmission will provide new openings and opportunities for industries and businesses, intensify digitisation, induce innovation unlocking the next level of growth, and boost the country's economy.

The list of new exciting and futuristic applications is long and the possibilities are enormous.

High-quality education

Use of the enhanced mobile broadband (eMBB) will spur rapid advancements in the field of high-quality digital education and skill development, delivering content through mobile applications to every student, and providing a major impetus to digital universities. Vocational training programmes in the phygital (integrated physical and digital) mode will improve the employability of youth and women.

Healthcare for all

With diagnostic data such as CT, MRI, and ultrasound becoming available in the cloud, it will be possible for the images to be fused instantly, making the scanning machines much simpler and cheaper. Also, the data on the cloud is easily accessed by doctors from anywhere, anytime. The technology will enable



5G enabled ML models are transforming medical diagnostics like detection of pancreatic cancer on pre-diagnostic CT scans (Source: <https://healthitanalytics.com>)

doctors arrive at decisions quickly, precisely, and accurately.

With easy accessibility to remote interventional support, wearable telemetry, transmission of HD images and electronic health records, hospitals will be able to provide quality care to large number of people with very few doctors and health care staff. The ultra-reliable low-latency communication (URLLC) and HD multimedia content will make possible point-of-care diagnostics, vitals monitoring, remote surgery, and connected ambulances.

NextGen banking

EmBB and URLLC with geospatial information systems (GIS) will make available simple, seamless, and secure payment methods, such as 'one-tap payment' and 'cashless store.' Mobile banking systems, such as virtual branches, will enable users in far flung areas to securely access banking facilities.

AR and VR in 'smart banks' will allow online facial

recognition, enabling customers to interact with bank officials. ATMs will offer AI-backed instant personalised services. Portable, pop-up virtual branches will become a possibility in remote locations or disaster-affected areas.

Efficient transportation

Telematics, digital cockpit, automatic driver assistance systems (ADAS), and cellular vehicle-to-everything communication is expected to considerably transform in-vehicle experience. Intelligent transportation will be possible using the autonomous driving, connected vehicles, and fleet management.

Integration of FASTags will improve the efficiency as well as help reduce carbon footprints. Massive machine-type communication (mMTC) will help optimise the usage of EVs, while making charging stations network cost effective. In ports, it will be possible to reduce inventory congestion using mMTC and URLLC.

Software-enabled automatic-guided vehicles will improve the port-space management while reducing the long waiting time. For the drones-as-a-service ecosystem,



A typical smart bank's self-service area (Source: <https://www.sedco.co>)